

12.744

EE25BTECH11002 - Achat Parth Kalpesh

October 11,2025

Question

Consider the following equations of straight lines:

$$\text{Line L1: } 2x - 3y = 5$$

$$\text{Line L2: } 3x + 2y = 8$$

$$\text{Line L3: } 4x - 6y = 5$$

$$\text{Line L4: } 6x - 9y = 6$$

Which one among the following is the correct statement?

- ① L1 is parallel to L2 and L1 is perpendicular to L3
- ② L2 is parallel to L4 and L2 is perpendicular to L1
- ③ L3 is perpendicular to L4 and L3 is parallel to L2
- ④ L4 is perpendicular to L2 and L4 is parallel to L3

Solution

A line $ax + by = c$ can be written as

$$\mathbf{n}^T \mathbf{x} = c, \quad (1)$$

where

$$\mathbf{n} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (2)$$

Thus, the normal vectors are:

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{n}_3 = \begin{pmatrix} 4 \\ -6 \end{pmatrix}, \quad \mathbf{n}_4 = \begin{pmatrix} 6 \\ -9 \end{pmatrix} \quad (3)$$

Solution

Two lines are parallel if their normal vectors are proportional:

$$\mathbf{n}_3 = 2\mathbf{n}_1 \implies \text{L1 is parallel to L3,} \quad (4)$$

$$\mathbf{n}_4 = \frac{3}{2}\mathbf{n}_3 \implies \text{L3 is parallel to L4.} \quad (5)$$

Two lines are perpendicular if

$$\mathbf{n}_i^T \mathbf{n}_j = 0 \quad (6)$$

$$\mathbf{n}_1^T \mathbf{n}_2 = \begin{pmatrix} 2 & -3 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 6 - 6 = 0 \implies \text{L1 is perpendicular to L2, (7)}$$

$$\mathbf{n}_4^T \mathbf{n}_2 = \begin{pmatrix} 6 & -9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 18 - 18 = 0 \implies \text{L4 is perpendicular to L2} \quad (8)$$

L4 is perpendicular to L2 and L4 is parallel to L3

C code

```
#include <stdio.h>
// Functions for each line:  $y = f(x)$ 
double line_L1(double x) {
    return (2.0*x - 5.0)/3.0;
}

double line_L2(double x) {
    return (8.0 - 3.0*x)/2.0;
}

double line_L3(double x) {
    return (4.0*x - 5.0)/6.0;
}

double line_L4(double x) {
    return (6.0*x - 6.0)/9.0;
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, c_double

# Load shared library
lib = CDLL("./formula.so")

# Define Python wrappers for C functions
lib.line_L1.restype = c_double
lib.line_L1.argtypes = [c_double]

lib.line_L2.restype = c_double
lib.line_L2.argtypes = [c_double]

lib.line_L3.restype = c_double
lib.line_L3.argtypes = [c_double]
```

Python Code

```
lib.line_L4.restype = c_double
lib.line_L4.argtypes = [c_double]

# x values for plotting
x = np.linspace(-5, 5, 400)

# Compute y values by calling C functions
y1 = np.array([lib.line_L1(xi) for xi in x])
y2 = np.array([lib.line_L2(xi) for xi in x])
y3 = np.array([lib.line_L3(xi) for xi in x])
y4 = np.array([lib.line_L4(xi) for xi in x])

# Plot each line
plt.plot(x, y1, label='L1:  $2x - 3y = 5$ ', color='blue')
plt.plot(x, y2, label='L2:  $3x + 2y = 8$ ', color='red')
plt.plot(x, y3, label='L3:  $4x - 6y = 5$ ', color='green')
plt.plot(x, y4, label='L4:  $6x - 9y = 6$ ', color='orange')
```

```
# Highlight parallel and perpendicular
plt.text(1, y1[240]+0.5, 'L1 || L3', color='green')
plt.text(2, y2[260], 'L2 $\perp$ L4', color='red')

# Axes and grid
plt.axhline(0, color='black')
plt.axvline(0, color='black')
plt.grid(True)
plt.legend()
plt.title("Visualization of Lines L1, L2, L3, L4")
plt.xlabel("X - axis")
plt.ylabel("Y - axis")
plt.xlim(-5, 5)
plt.ylim(-5, 5)
plt.show()
```

Python Plot

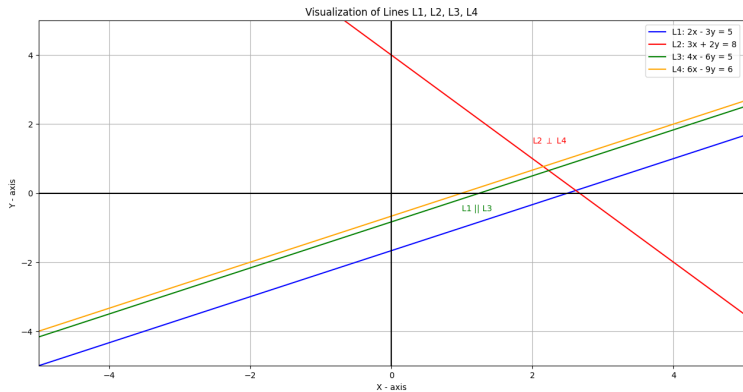


Figure: Lines